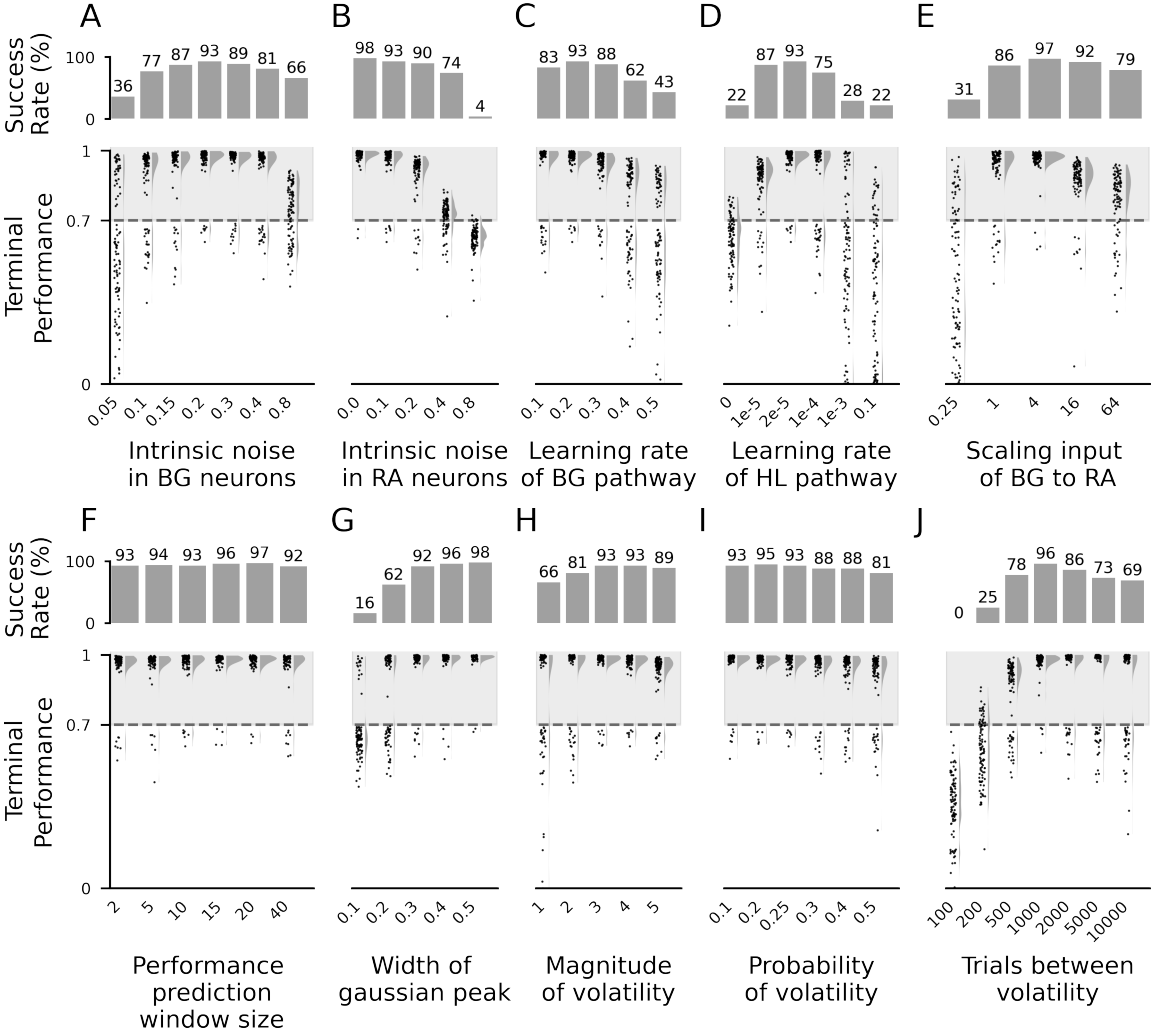

A
B
C
D
E
F
G
H
I
J
Success Rate (%)
Terminal Performance
100
0
1
0.7
0
36 77 87 93 89 81 66
98 93 90 74 4
83 93 88 62 43
22 87 93 75 28 22
31 86 97 92 79
93 94 93 96 97 92
16 62 92 96 98
66 81 93 93 89
93 95 93 88 88 81
0 25 78 96 86 73 69
0.05 0.1 0.15 0.2 0.3 0.4 0.8
Intrinsic noise in BG neurons
0.0 0.1 0.2 0.4 0.8
Intrinsic noise in RA neurons
0.1 0.2 0.3 0.4 0.5
Learning rate of BG pathway
0 1e-5 2e-5 1e-4 1e-3 0.1
Learning rate of HL pathway
0.25 1 4 16 64
Scaling input of BG to RA
2 5 10 15 20 40
Performance prediction window size
0.1 0.2 0.3 0.4 0.5
Width of gaussian peak
1 2 3 4 5
Magnitude of volatility
0.1 0.2 0.25 0.3 0.4 0.5
Probability of volatility
100 200 500 1000 2000 5000 10000
Trials between volatility